

Generation Interconnection System Impact Study Report

For Queue Project Q24-2382

Ss3145Eagle Rock Tap 1 345 kV
December 24, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2382. The Interconnection Customer proposes a 30.0 MW solar photovoltaic facility interconnecting at the Ss3145Eagle Rock Tap 1 345 kV substation in the Kentucky area. The request was received on February 14, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 338 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,587,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- 1. Facility: Rancho Carmel Substation / Russellville East 138 kV Line
Contingency: Berry Street Substation / Northfield 345 kV
Loading: 120.2% (Rating: 414 MVA)
DFAX: 8.8%
Pre-project loading: 119.5%
- 2. Facility: Sand Springs / Russellville East 138 kV
Contingency: Berry St / Northfield 345 kV
Loading: 112.7% (Rating: 434 MVA)
DFAX: 8.8%
Pre-project loading: 112.1%
- 3. Facility: Frontier / Delano 138 kV
Contingency: Mcclellan Ar / Spatsum Lornex 138 kV
Loading: 112.3% (Rating: 202 MVA)
DFAX: 29.6%
Pre-project loading: 107.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rancho Carmel / Russellville E 138 kV Line	Rebuild line with higher-capacity conductor	\$12,125,000	1% (\$160,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Russellville E / Sand Springs 138 kV	Rebuild line with higher-capacity conductor	\$8,664,000	1% (\$114,000)
Frontier / Delano Substation 138 kV Line	Rebuild line with higher-capacity conductor	\$7,943,000	16% (\$1,313,000)
	Total allocated to project		\$1,587,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.